

SECTION A
TRUE OR FALSE STATEMENTS

[10 marks: 1 mark for a correct answer and - ½ for an incorrect answer]

1. Since the 1700s, increases in average living standards became a permanent feature of economic life in many countries. This process has been accompanied by growing threats to our natural environment.
a) True
b) False
2. A firm uses labour and coal to produce a pair of shoes. Coal costs R12.50 per ton and labour costs R5 per worker per hour. On the firm's R300 isocost line, the firm will give up 0.8 tons of coal to hire two additional workers.
a) True
b) False
3. Alicia consumes only sour worms and magazines. A change in Alicia's income will change the shape of her indifference curves.
a) True
b) False
4. Suppose that a standard Ultimatum game is adapted so that the Proposer makes an offer to a group of four Responders. As long as the offer is accepted by at least one of the Responders, the split will be implemented. Under this variation of the Ultimatum game, Responders will be more likely to accept lower offers made by the Proposer than would be the case in a standard Ultimatum game.
a) True
b) False
5. If Bruno can coerce Angela to work on his land, Bruno is indifferent between all allocations that lie on Angela's biological survival constraint.
a) True
b) False
6. In economics, we use the term “employment rent” to refer to the wage workers earn.
a) True
b) False
7. In a perfectly competitive market, the market price is equal to the marginal cost of production.
a) True
b) False

8. A fall in South Africans' incomes due to unemployment caused by Covid-19-related restrictions leads to a leftward shift of the demand curve for houses.
- a) True
 - b) False
9. Monopolies keep prices high by producing a quantity that is higher than the Pareto-efficient allocation.
- a) True
 - b) False
10. Pure public goods are non-rival and non-excludable.
- a) True
 - b) False

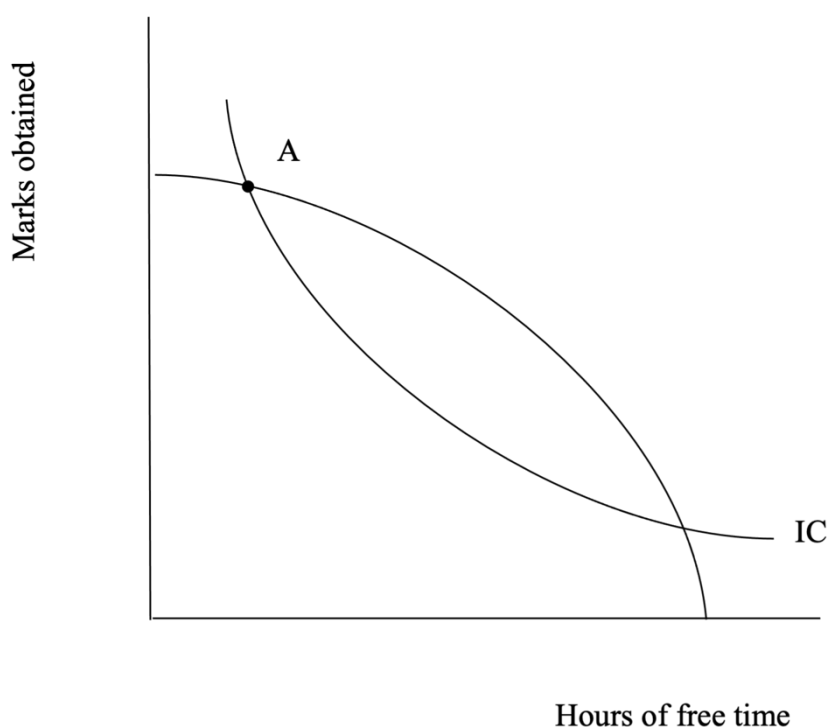
SECTION B

MULTIPLE CHOICE QUESTIONS

[30 marks: 2 marks for a correct answer and – ¼ for an incorrect answer]

11. Capitalism is an economic system in which private property, markets, and firms play an important role. Given this definition, which one of the following statements is **CORRECT**?
- a) An economic system is a way of organising the production and distribution of goods and services in an entire economy.
 - b) The knowledge you attain from the Economics 114 module is considered private property.
 - c) There is no role for governments in modern capitalist systems.
 - d) Firms in capitalist systems are usually managed by the workers.
12. Which of the following statements is/are **CORRECT**?
- i. Technological advancement and the division of labour have both played an important role in increasing the scale of production of firms in a market economic system.
 - ii. Firms competing with one another provide an incentive to refrain from adopting and developing new and more productive technologies.
 - iii. It is impossible for markets to exist outside a capitalist system.
 - iv. Some of the ideas about capitalism can be attributed to Adam Smith.
- a) Only i is correct.
 - b) ii and iii are correct.
 - c) i and iv are correct.
 - d) i, iii and iv are correct.

13. Suppose that the price of good Y decreases, while the price of good X and a person's income remain the same. If the person's consumption of X remains unchanged after the decrease in the price of Y, it means that:
- a) the income and substitution effects are not applicable in this instance.
 - b) the income and substitution effects on the consumption of X are zero.
 - c) the increase in the consumption of X (caused by the income effect) is countered exactly by the decrease in consumption of X (caused by the substitution effect).
 - d) the income and substitution effects work in the same direction.
14. The figure below shows the feasible frontier and indifference curve of a high school student who faces a tradeoff between free time and school marks obtained.



At point A the student is not in equilibrium because _____. He could increase his utility by _____.

- a) $MRT > MRS$; increasing his free time
- b) $MRT > MRS$; studying more
- c) $MRT < MRS$; studying more
- d) $MRT < MRS$; increasing his free time

15. The production function for sneakers is given by:

$$f(h) = 4h^2 + 6h + 20$$

The marginal product of labour in the factory is given by:

- a) $4h + 6$
b) $8h + 6h + 20$
c) $8h + 6$
d) 20
16. Palesa and her best friend, Thandi, want to catch up but cannot do so in person as a result of the Covid-19 social distancing regulations. They can meet via WhatsApp or FaceTime, and their interaction is represented by the game in the table below:

		Thandi	
		WhatsApp	FaceTime
Palesa	WhatsApp	3, 3	1, 0
	FaceTime	0, 1	2, 2

Which one the following statements is **INCORRECT**?

- a) When Palesa plays WhatsApp, Thandi's best response is WhatsApp.
b) (WhatsApp, WhatsApp) and (Facetime, Facetime) are Nash equilibria in this game.
c) There is no dominant strategy equilibrium in this game.
d) Thandi has a dominant strategy to play WhatsApp.
17. Which one of the following is a procedural judgement of fairness?
- a) Wealth should be distributed as equally as possible.
b) Politicians should not enrich themselves through corruption.
c) Everyone should be able to satisfy their basic needs.
d) Celebrities do not deserve to earn more than ordinary people.
18. Evaluate the following two statements to answer this question:
- i. The principal-agent problem can be explained by a lack of incentives.
ii. The key problem in principal-agent situations is the manager's inability to supervise the actions of the worker.
- a) i and ii are both true.
b) i is true and ii is false.
c) ii is true and i is false.
d) i and ii are both false.

19. Which one of the following statements regarding employment contracts is **CORRECT**?
- The firm is required to state exactly what it needs the employee to do in an employment contract.
 - The firm needs to specify exactly how much effort employees are expected to put into their job.
 - Employees' effort levels cannot be the basis of an enforceable contract.
 - Employment contracts are incomplete as they can only specify things that both the employees and the business owner care about.
20. In the market for premium headphones, the market demand is $P = 5000 - 10Q$, while the market supply is $P = 500 + 5Q$. Suppose the government places a tax on the production of headphones of R450 per unit. Which of the following statements are **CORRECT**?
- 300 headphones will be traded after the introduction of the tax.
 - 270 headphones will be traded after the introduction of the tax.
 - The price of headphones will increase by R450.
 - The price of headphones will increase by R300.
 - Producers bear a greater share of the tax burden than consumers.
- i and iii
 - iii and v
 - ii and iv
 - ii, iii and v
21. Which one of the following statements regarding firms with market power is **CORRECT**?
- Firms with market power cannot influence the price that is set in the market.
 - Firms with market power produce more goods than firms without market power, *ceteris paribus*.
 - Marginal revenue is always constant for firms with market power.
 - Some firms with market power are able to make economic profits in the long run because barriers to entry prevent competition from emerging.
22. The price of insulin (a medicine used by diabetics) increases but the quantity of insulin purchased by diabetics remains the same. This is an example of:
- price elastic demand
 - perfectly price inelastic demand
 - unitary price elasticity of demand
 - perfectly price elastic demand

23. In the market for budget laptops, the effect of an increase in the price of premium laptops (a substitute for budget laptops) along with an increase in the cost of raw materials (used in the production of budget laptops) will cause _____ in the equilibrium price and _____ in the equilibrium quantity in the market for budget laptops.
- a) an increase; an increase
 - b) an increase; a decrease
 - c) a decrease; an indeterminate change
 - d) an increase; an indeterminate change
24. In the case of a negative externality, there is _____ and _____.
- a) underproduction; $MSC > MSB$
 - b) underproduction; $MSB > MSC$
 - c) overproduction; $MSC > MSB$
 - d) overproduction; $MSB > MSC$

25. The consumption of fairy dust causes positive externalities. Consider the following equations:

$$MC = 5500 + Q$$

$$MPB = 8000 - 2Q$$

$$MEB = 1000 - Q$$

What subsidy per unit should the fairy queen pay to the fairy dust industry to arrive at the socially optimal outcome?

- a) 125
- b) 875
- c) 6 250
- d) 6 375

SECTION C
DESCRIPTIVE QUESTIONS
[60 marks]

QUESTION 26 [2 marks]

The GDP per capita of Greece was \$22 494 in 2021 and \$21 966 in 2022. Based on these figures, calculate the growth rate of GDP between 2021 and 2022. Round your answer off to two decimal places. [2]

QUESTION 27 [5 marks]

- 27.1 Living standards remained at or near subsistence levels for centuries before the Industrial Revolution. Briefly explain this phenomenon using Malthus's theory. [3]
- 27.2 Name two conditions that allowed countries to escape from the Malthusian trap. [2]

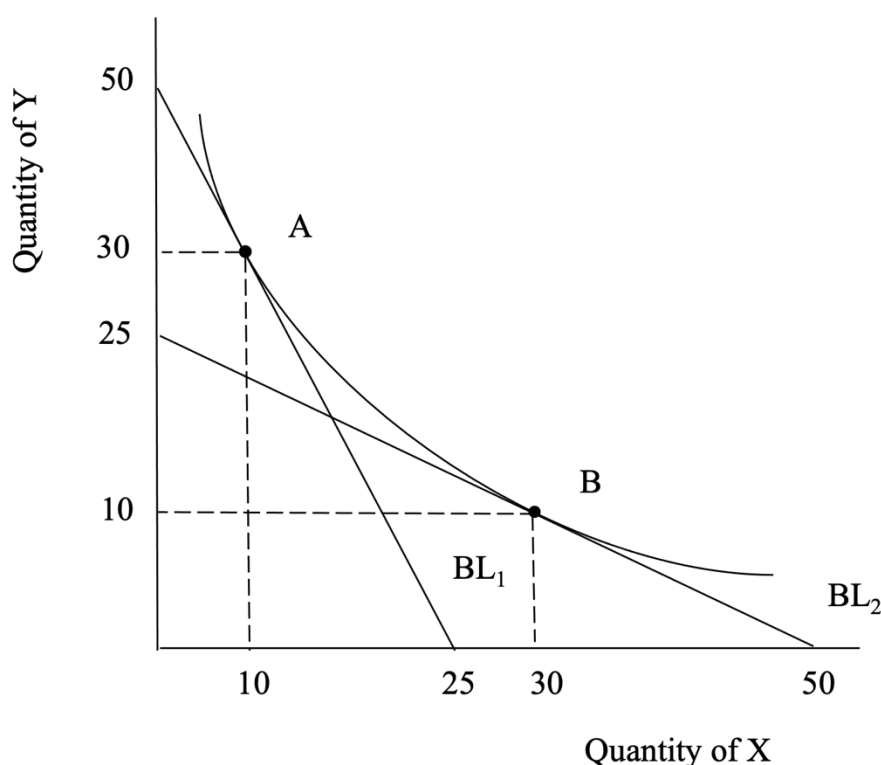
QUESTION 28 [6 marks]

A worker faces a consumption constraint $c = 10(24 - t)$, where t is hours of free time per day and c is consumption. The worker is paid in Rands.

- 28.1 Assume that this worker can work for a maximum of 15 hours a day. What is the worker's maximum possible daily consumption? [2]
- 28.2 Assume that an anonymous benefactor pays this worker R100 every day, irrespective of the hours worked. What is the opportunity cost of one hour of free time? [1]
- 28.3 The worker is now told that his wage is going to increase by R5 for every hour worked. He will still receive the R100 from the benefactor.
- 28.3.1 Write an equation for his new consumption constraint. [1]
- 28.3.2 What is the substitution effect of this wage change on free time? Explain your answer. No calculations are required. [2]

QUESTION 29 [5 marks]

The diagram below shows the budget constraints before and after a change in prices and income, as well as an indifference curve of a consumer of Good X and Good Y.



- 29.1 Calculate the value of the MRS at point A. [2]
- 29.2 On BL_1 , if the price of Good X is R4, what is

29.2.1 the price of Good Y? [1]

29.2.2 the consumer's income? [1]

29.3 On BL₂, what is the relative price of Good Y? [1]

QUESTION 30 [3 marks]

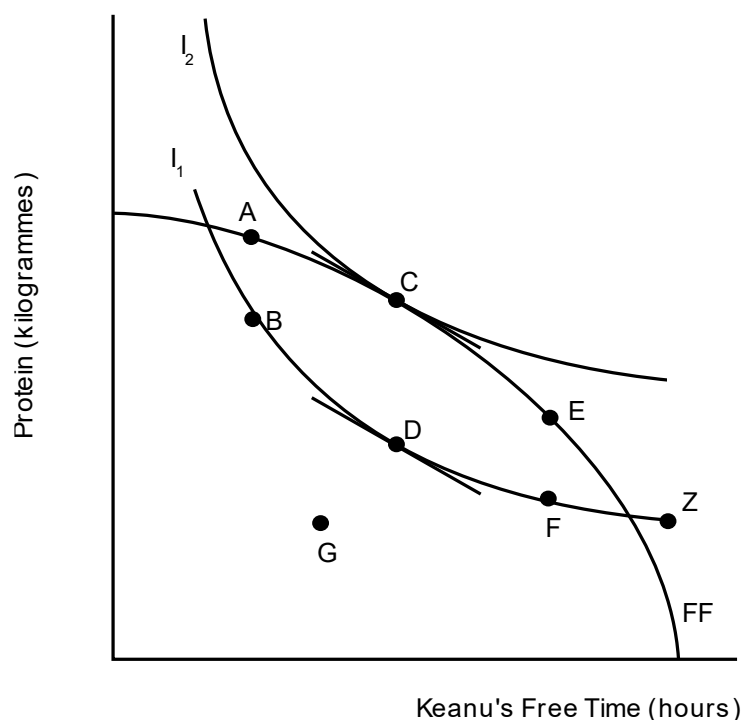
Sheldon and Leonard are splitting housework. The social interaction of washing dishes is represented in the table below:

		Leonard	
		Wash dishes	Don't wash dishes
Sheldon	Wash dishes	12, 12	2, 15
	Don't wash dishes	15, 2	5, 5

What is the Nash equilibrium of this game and explain why it is a social dilemma. [3]

QUESTION 31 [5 marks]

Keanu works in Wachowski's factory, producing Protein, which both Keanu and Wachowski consume. The graph below shows the feasible frontier (FF) which indicates combinations of Keanu's free time and Protein production that is feasible to produce. Wachowski consumes the Protein that is produced by Keanu that Keanu does not consume himself. Point Z represents Keanu's reservation option. Production will only occur if a voluntary agreement can be reached between Keanu and Wachowski. Note that the two straight lines through C and D on the graph are parallel.



- 31.1. Which point (A, B, C, D, E, F, G or Z) would be Wachowski's most preferred on the graph? Briefly motivate your answer. [2]
- 31.2. Suppose Wachowski and Keanu bargain with each other until a voluntary agreement is reached, and that they have roughly equal bargaining power. Where on the graph would the outcome be? Motivate your answer. [3]

QUESTION 32 [4 marks]

Aniya earns R15 per hour in her current job and works 40 hours a week. Her disutility of effort is measured as R1 per hour of work. In the case that she loses her job, she receives R5 per hour unemployment benefit. Additionally, being unemployed has psychological and social costs which is a disutility equivalent to R2 per hour. Calculate Aniya's total employment rent if she can get another job with the same wage rate after 44 weeks of being unemployed. [4]

QUESTION 33 [11 marks]

Julian's Chesa-Nyama is one of many businesses operating in a competitive braai food market in Stellenbosch. The inverse demand function for braai meals is represented by:

$$P = 68 - 0.02Q$$

where Q represents the number of plates of braai meals sold in Stellenbosch per day. The inverse supply function for braai meals is represented by:

$$P = 28 + 0.005Q$$

33.1 Determine the market equilibrium price and quantity of braai meals. [3]

33.2 What is the price elasticity of demand for Julian's braai meals? [1]

Julian's total cost of production is given by:

$$TC = 40 + 28q + \frac{1}{8}q^2$$

where q represents the number of plates of braai meals sold by Julian per day.

33.3 How many braai meals should Julian sell per day to maximise his economic profit? [3]

33.4 Calculate Julian's economic profit. [2]

33.5 Can Julian increase his profit by raising the price of his meals? Why or why not? [2]

QUESTION 34 [4 marks] Lindi

The equations below describe the market demand and supply curves of sneakers:

$$P = 32 - Q_d$$

$$P = Q_s + 8$$

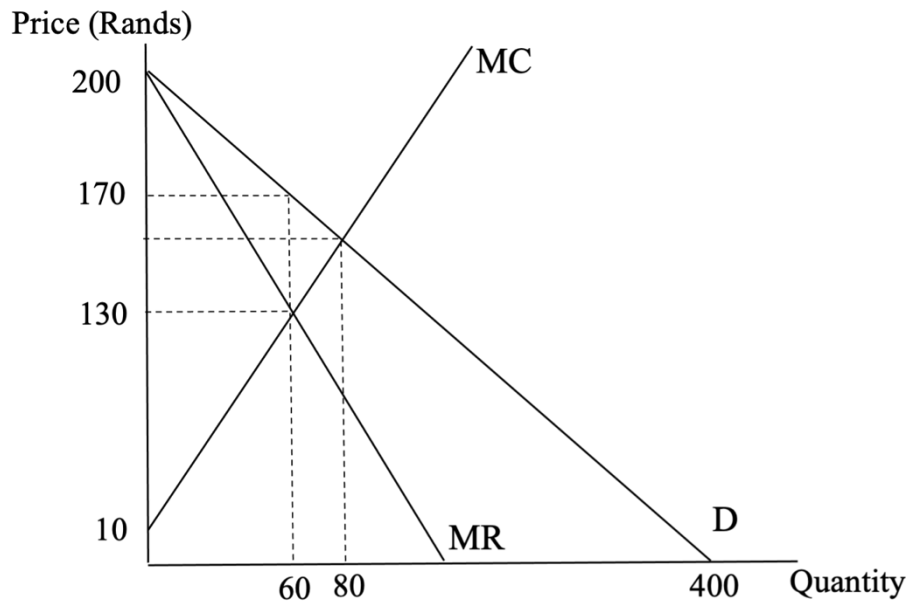
Assume that the government imposes a price ceiling of R11 per pair of sneakers. Answer the following questions that refers to the period after the price ceiling was introduced.

34.1 Is there a shortage or a surplus in this market? What is the size thereof? [3]

34.2 How does the introduction of a price ceiling affect the size of the consumer surplus? No calculations are required. [1]

QUESTION 35 [9 marks] Lindi

The figure below indicates the demand, marginal revenue, and marginal cost curves of a monopoly.



- 35.1 Write down the equations for the demand and marginal cost curves. [2]
- 35.2 Calculate the producer and consumer surplus. Show all calculations. [4]
- 35.3 Calculate the price mark-up. [1]
- 35.4 Calculate the size of the deadweight loss. [2]

QUESTION 36 [6 marks]

The Flotsams like to have loud parties at night, while their neighbours the Jetsams suffer harm from the noise. The table below shows the marginal private cost (MPC) and marginal private benefit (MPB) of parties to the Flotsams, as well as the marginal external cost (MEC) of the parties. Show all calculations for the questions below.

Q	MPB	MPC	MEC
1	101	40	40
2	91	50	40
3	81	60	40
4	71	70	40
5	61	80	40
6	51	90	40
7	41	100	40

- 36.1 How many parties per month will the Flotsams choose if they can freely choose without legal or social restrictions? (Note, they can only choose an integer number.) [1]
- 36.2 What is the socially optimal number of parties? [1]
- 36.3 How much will it benefit the Jetsams if the number of parties were reduced to the socially optimal level? [2]
- 36.4 If property rights were well defined, what private agreement could be made between the Flotsams and the Jetsams? [2]